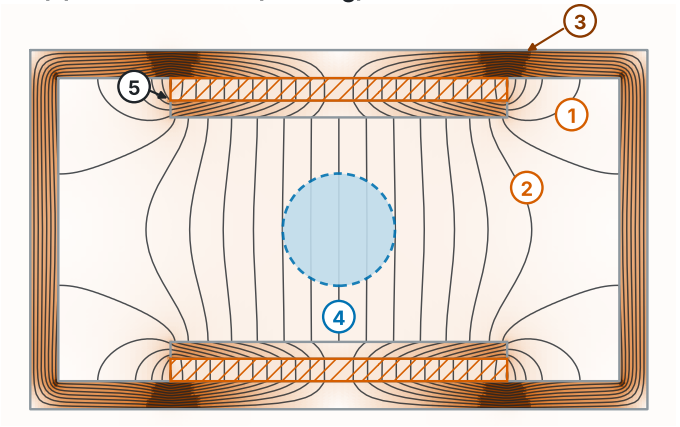
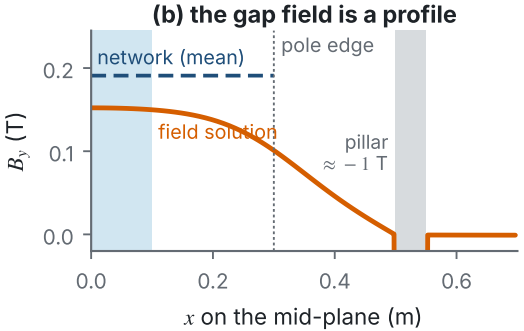
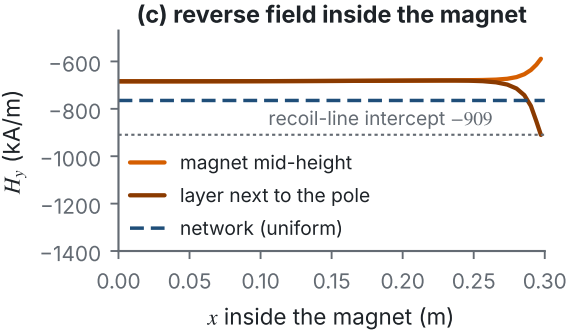

(a) 2-D section: |B| (shading) and flux lines
3
5
1
2
4
1 leakage: only 46% of the magnet flux crosses between the poles
2 pole edge: lines bulge, field drops
3 local saturation: plate > 1.5 T (dark)
4 imaging region r = 0.1 m: ΔB/B = 2.5%
5 magnet corner: strongest reverse field
the network gives one mean B_g only


(b) the gap field is a profile
network (mean)
pole edge
field solution
pillar
≈ − 1 T
B_y (T)
0.2
0.1
0.0
0.0
0.2
0.4
0.6
x on the mid-plane (m)


(c) reverse field inside the magnet
recoil-line intercept −909
magnet mid-height
layer next to the pole
network (uniform)
H_y (kA/m)
−600
−800
−1000
−1200
−1400
0.00
0.05
0.10
0.15
0.20
0.25
0.30
x inside the magnet (m)